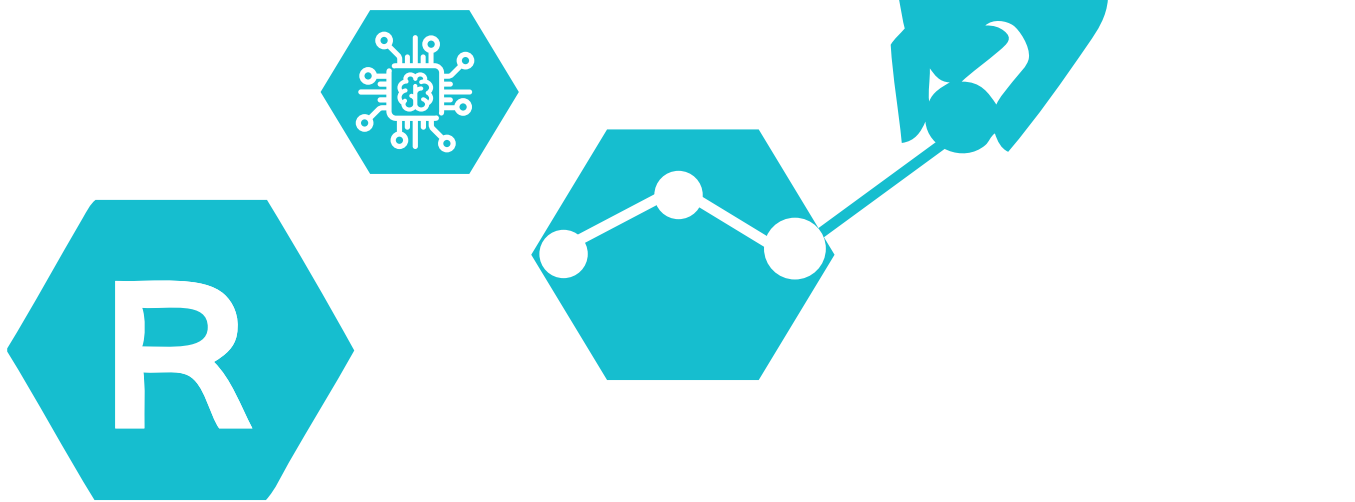




An Introduction to R

Welcome to a new series on 'R, statistics and machine learning'. R is a programming language that was primarily designed for statistical computing and graphics. It is a multi-paradigm programming language that supports an imperative, object-oriented, array and functional style of programming. R is dynamically typed and is primarily written in C, Fortran and R itself.

R is an official GNU package and is released under the GNU GPL v2 licence. It was first released in 1993 and the latest stable release is 4.0.4. The official home page of the R project is <https://www.r-project.org/>. In this new series of articles, we will explore the syntax, semantics of R and also the various libraries available for statistics, graphics and machine learning.

Installation

Parabola GNU/Linux-libre: You can install R on Parabola GNU/Linux-libre using the Pacman package manager, as shown below:

```
$ sudo pacman -S r
```

The latest version that gets installed is 4.0.4-1, as indicated below:

```
extra/r 4.0.4-1 [installed]
  Language and environment for statistical computing
  and graphics
```

Debian/Ubuntu: The 'r-base' package needs to be installed on Ubuntu to get R in your system:

```
$ sudo apt install r-base
```

Fedora: The latest R version can be installed on Fedora using:

```
$ sudo dnf install R
```

Mac OS X: The 'R.APP' application can be installed from <https://mac.r-project.org/> for Mac OS X. The website provides both the *-devel* and *-stable* releases for installation. Periodic nightly builds are made for the R releases with a .pkg file. Please note that these releases for Mac OS X are still experimental in nature.

Windows: The 'bin/windows/base' directory at <https://cran.r-project.org/mirrors.html> provides an *R-4.0.4-win.exe* executable for R on Windows. If you like to test the latest software, you can install the 'r-patched' or 'r-devel' snapshot releases as well. R on Windows is supported from Windows 7 or later, and the installation takes at least 150MB of disk space.

Emacs: As an Emacs user, you can install the 'Emacs Speaks Statistics' (ESS) package that provides support for working on R source files. The add-on includes syntax highlighting, code formatting, searching for documentation, displaying results, etc. The project website is available at <https://ess.r-project.org/>. With a Cask setup, you can simply add the following to your Cask file to install ESS:

```
(depends-on "ess")
```